



Geometric Dictionary Learning of Dynamical Systems with Optimal Transport

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Dynamical system & machine learning

Dynamical systems are backbone models of temporally evolving phenomena:

Modeling:

$$x_{t+1} = g(x_t)$$

discrete

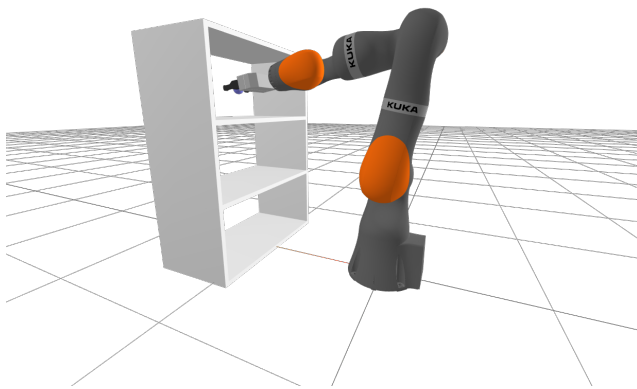
$$dx_t = g(x_t)$$

continuous

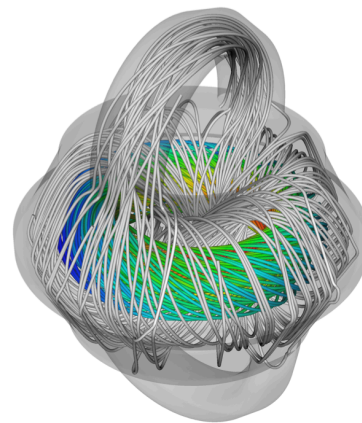
$$dX_t = g(X_t) + \sigma dB_t$$

stochastic

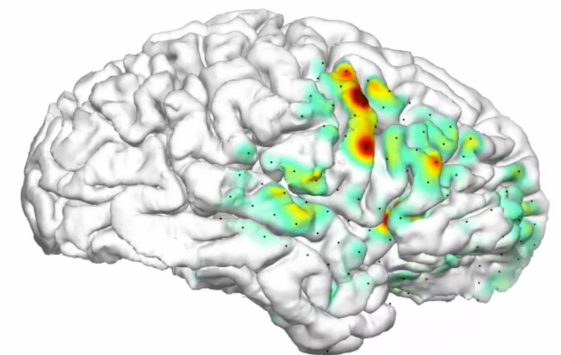
Application:



Robotic



Fluid dynamic



Brain dynamic

Machine learning for dynamical systems

Surrogate model & forecasting:

- ➡ Learning efficient models to predict future states (Neural operators).

Control & decision making:

- ➡ Understanding how interventions on state variables propagate through the system (causal inference).
- ➡ Learning control policies to drive system towards an objective (model-based RL)

System identification & model discovery:

- ➡ Learning the underlying transition equations (neuralODE)
- ➡ Learning coordinate systems in which dynamics become simple (Koopman theory)

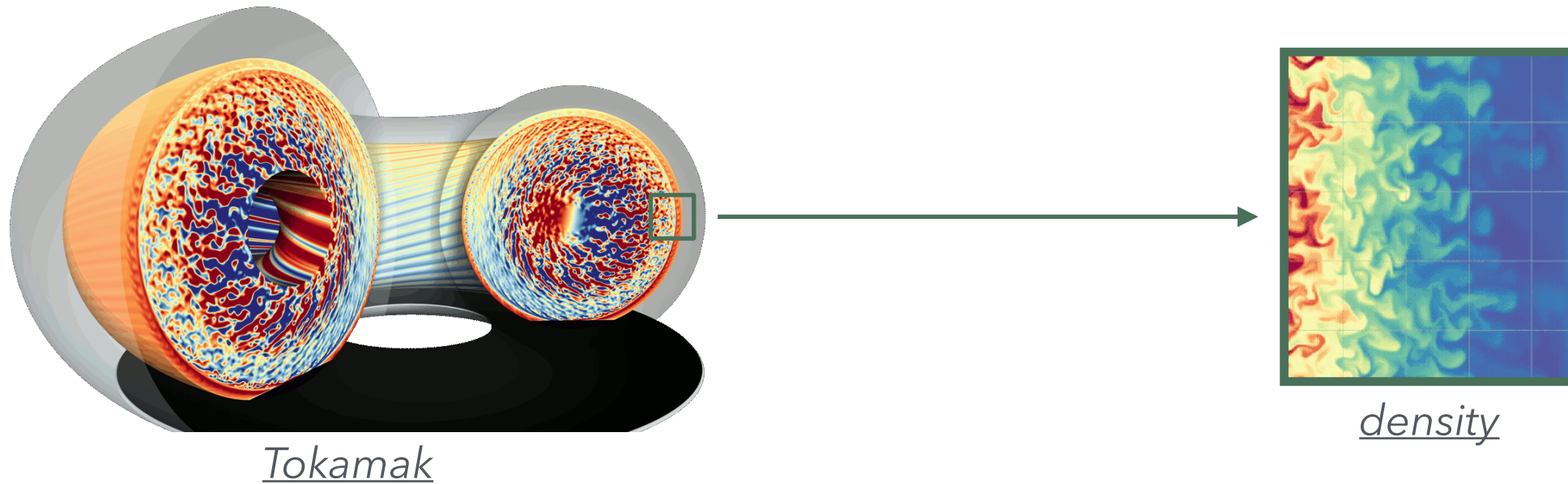


Current methods typically consider a single dynamic.

- ➡ Necessity of long trajectory (large dataset)

Can we leverage knowledge about « similar » dynamics to perform model discovery ?

Application: Plasma dynamic identification



Challenge: Heat and confine plasma long enough for fusion reactions.

Plasma PDE model:

$$\begin{cases} \partial_t n + \kappa \partial_y \phi + [\phi, n] - D_n \Delta_{\perp} n = -\sigma_n n + \sigma_{n\phi} \phi, \\ \partial_t W + g \partial_y n + [\phi, W] - D_{\phi} \Delta_{\perp} W = -\sigma_{\phi n} n + \sigma_{\phi\phi} \phi \\ W = \Delta_{\perp} \phi \end{cases}$$

with n the particle density, ϕ the electric potential and W the vorticity.

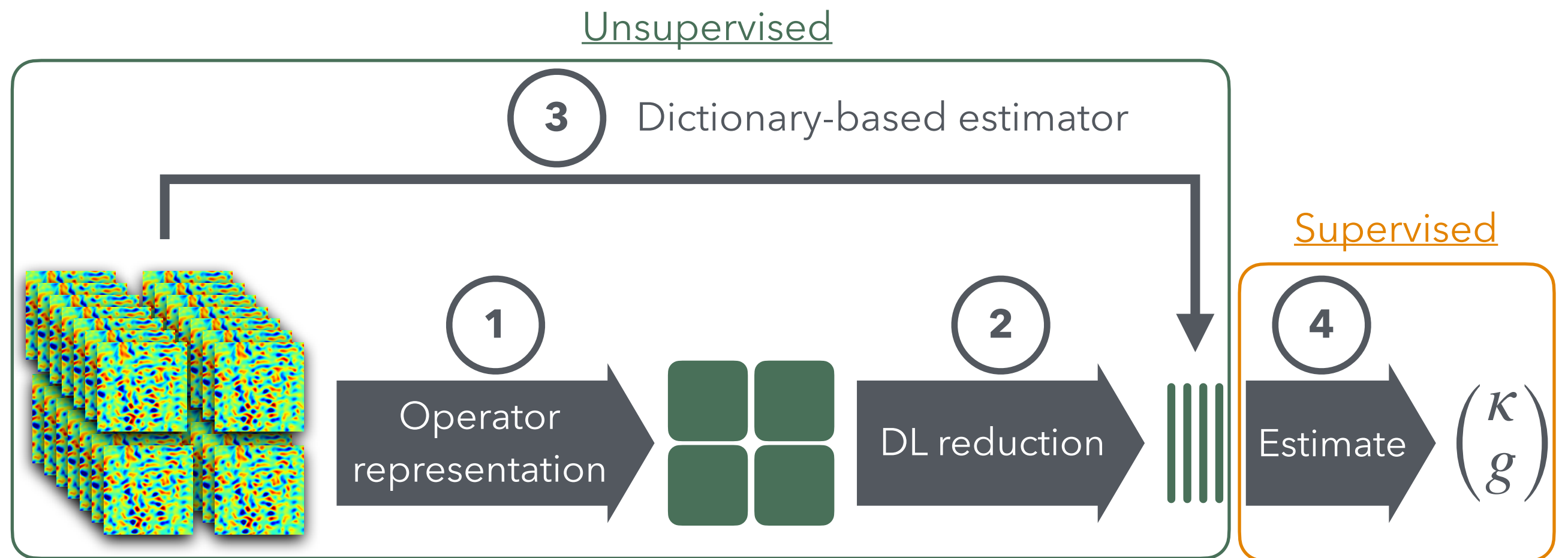
Intermediate challenge:

Can we estimate parameters κ and g ? Even from short trajectories ?

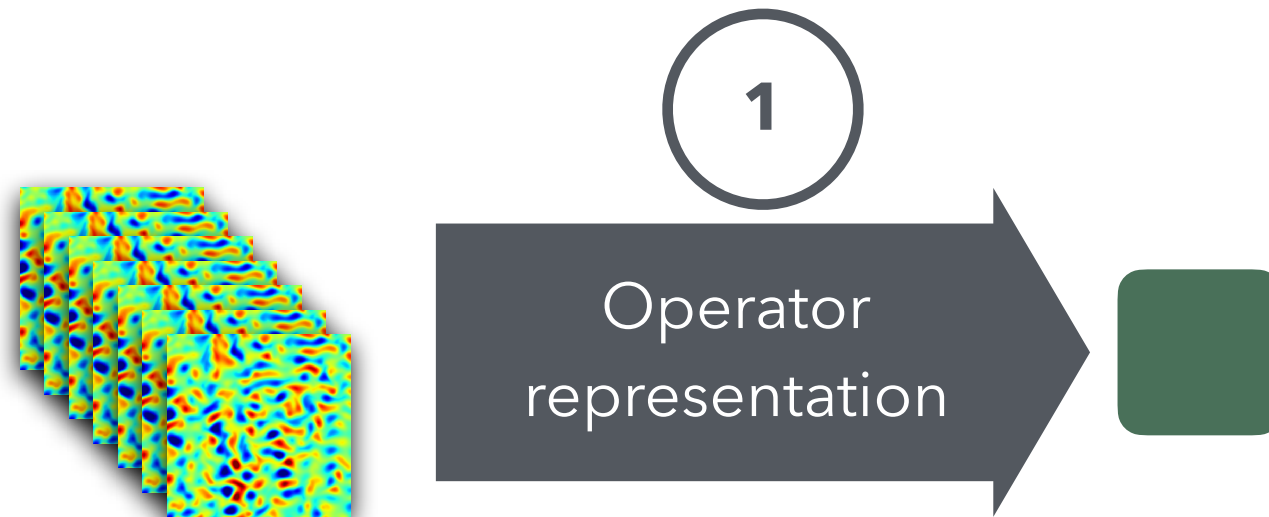
Overview

Developed methodology:

1. A pipeline to transform raw data into PDE's operator representation (matrix)
2. Compress representation with dictionary learning (DL) (matrix to small vector)
3. Dictionary-based estimation of operator representations.
4. Estimation of PDE's parameters from short trajectory.



Operator representation:



Main question:

Given a sampled trajectory, $\{x_t\}_{t \in [N]}$,
how to represent the underlying unknown dynamic ?

Linear autonomous dynamical systems

Modeling:

$$\dot{x}(t) = Ax(t) \rightarrow x(t) = \exp(tA)x_0$$

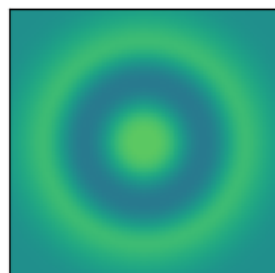
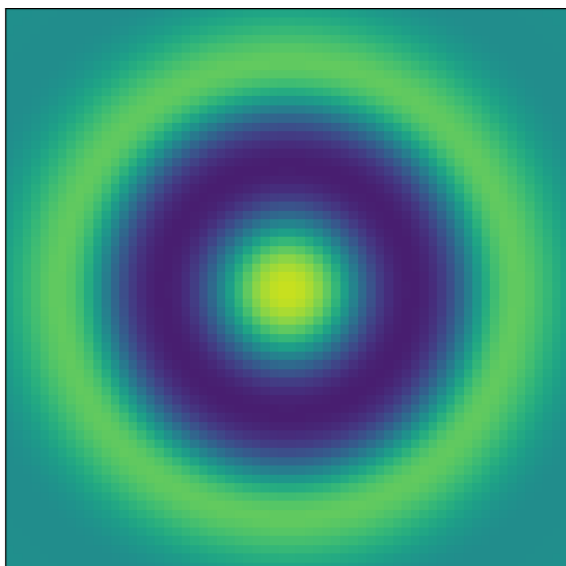
Non-defective system (spectral decomposition):

$$A = \underbrace{\begin{pmatrix} | & & | \\ p_1 & \cdots & p_n \\ | & & | \end{pmatrix}}_P \underbrace{\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}}_\Lambda \underbrace{\begin{pmatrix} - & p_1^{-1} & - \\ & \vdots & \\ - & p_n^{-1} & - \end{pmatrix}}_{P^{-1}}$$

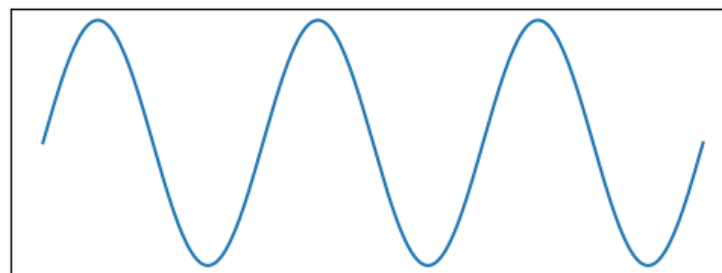
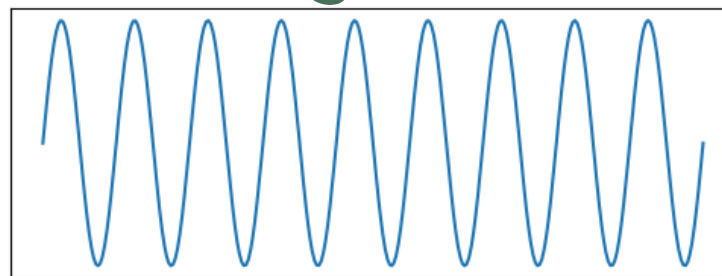
(Superposition principle)
$$x(t) = \sum_{k=1}^n \boxed{e^{t\lambda_k}} \boxed{p_k \langle p_k^{-1}, x_0 \rangle} = \sum_{k=1}^n \boxed{e^{t\tau_k + i2\pi\omega_k t}} \boxed{m_k(x_0)}$$

➔ Sum of elementary solutions that decouple **temporal** and **static** components.

Modes



Signals



- $\tau_k = \mathcal{R}(\lambda_k)$ timescale.
- $\omega_k = \mathcal{I}(\lambda_k)/2\pi$ frequency.
- $m_k(x_0) = \langle p_k^{-1}, x_0 \rangle p_k$ **mode**.

Non-linear autonomous dynamical systems

Modeling:

$$\dot{x}(t) = g(x(t)) \quad \rightarrow \quad x(t) = S_t(x_0) = x_0 + \int_0^t g(x(\tau)) d\tau$$

Koopman operator: Suppose a function space $\mathcal{F} \subset \mathbb{R}^{\mathcal{X}}$ (observables), for any time $t > 0$, the Koopman operator $T_t : \mathcal{F} \rightarrow \mathcal{F}$ associated to the dynamic verifies:

$$[T_t f](x_0) \triangleq f(S_t(x_0)) \quad \forall f \in \mathcal{F}, \forall x_0 \in \mathcal{X}$$

Remarks: Koopman operators are linear: $T_t(f + cg) = T_t f + cT_t g$

Infinitesimal generator: The semi-group of operators $\{T_t\}_{t>0}$ (i.e. $T_{s+t} = T_s T_t$) admits an infinitesimal generator such that:

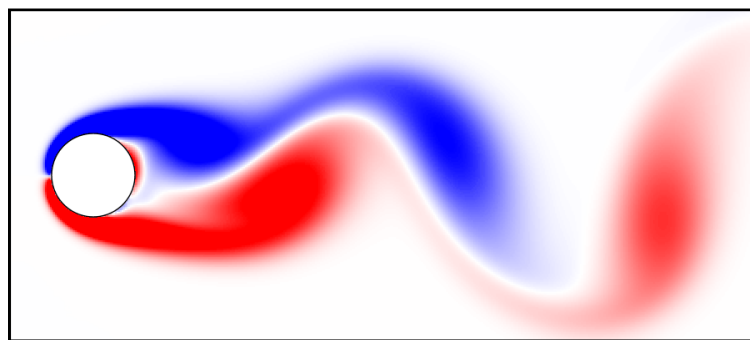
$$A = \lim_{t \rightarrow 0} (T_t - Id)/t \quad \text{and} \quad T_t = \exp(tA)$$

Operator spectral decomposition

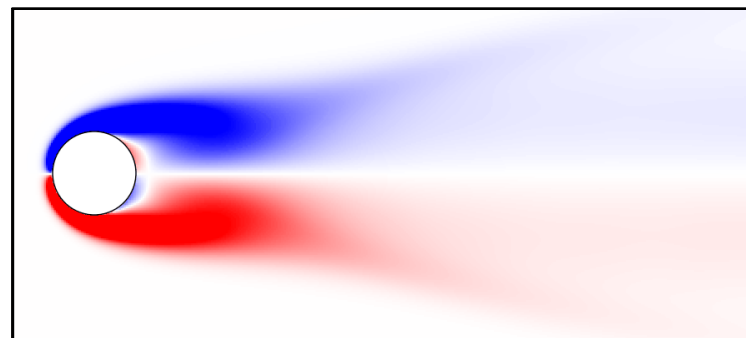
Spectral decomposition: Assuming that $\mathcal{F}$ is a separable Hilbert space (typically $L^2_\pi(\mathcal{X})$ with π the system's invariant measure) and A is a non defective operator with purely discrete spectrum. Then A can be written as:

$$Af = \sum_{j \in \mathbb{N}} \lambda_j \langle g_j, f \rangle f_j \quad \text{s.t.} \quad Lf_j = \lambda_j f_j, \quad L^*g_j = \bar{\lambda}_j g_j, \quad \langle f_i, g_j \rangle = \delta_{i,j}$$

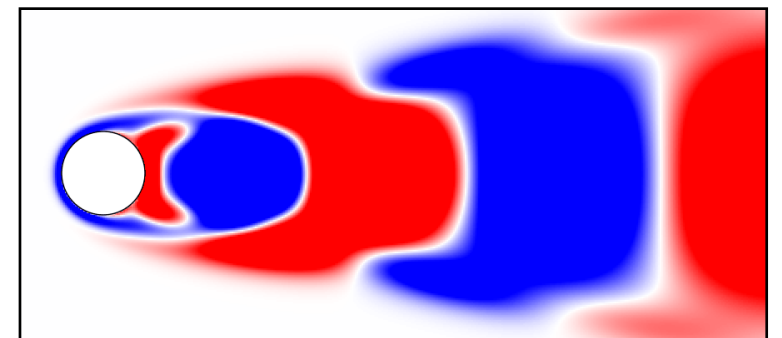
Raw signal



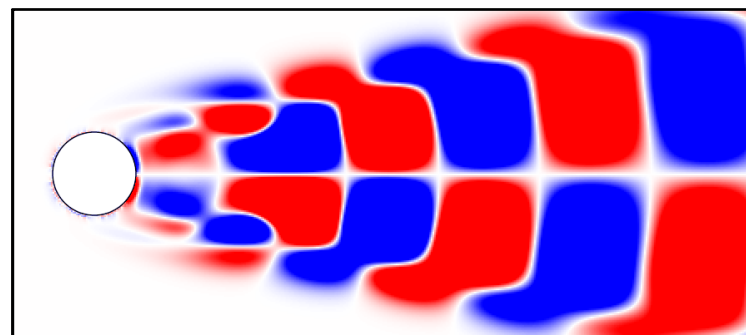
Mode 1 (0Hz)



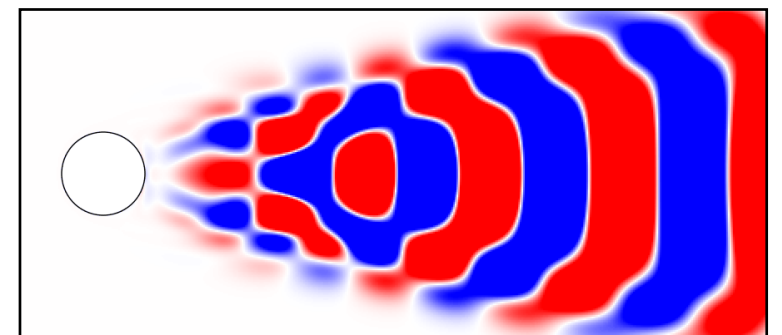
Mode 2 (3.3Hz)



Mode 3 (6.4Hz)

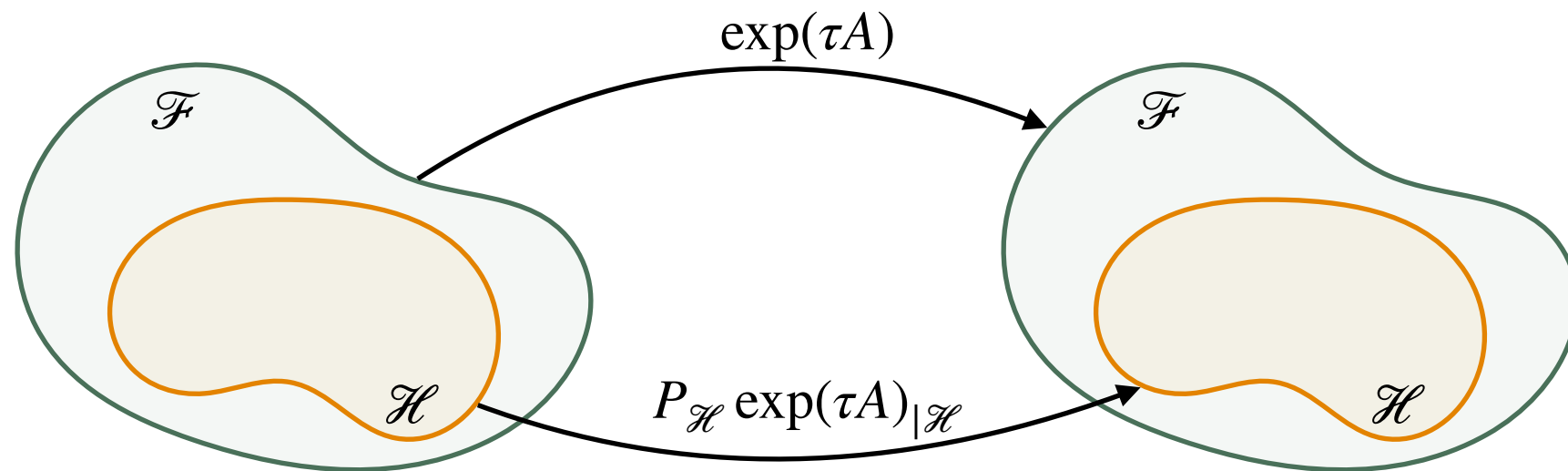


Mode 4 (9.3Hz)



Learning transfer operators

Problem: Only trajectories of dynamical systems are observed. Neither the operator A nor its domain $\mathcal{F}$ are known.



Popular approach [Bru22, Kos22]: Learning the action of $A : \mathcal{F} \rightarrow \mathcal{F}$ on a predefined finite dimensional RKHS $\mathcal{H} \subset \mathcal{F}$ with basis $(f_i)_{i \in [d]}$ and features map $\phi(x) = (f_1(x), \dots, f_d(x))$ via empirical risk minimization:

$$\operatorname{argmin}_{A \in \mathbb{R}^{r \times r}} \frac{1}{n} \sum_{t \in [n]} \|z_{t+1} - A^\dagger z_t\|_{\mathcal{H}}^2 \quad \text{s.t.} \quad z_t = \phi(x_t)$$

Dictionary learning embedding reduction



Main question:

How to compress operator representation ?

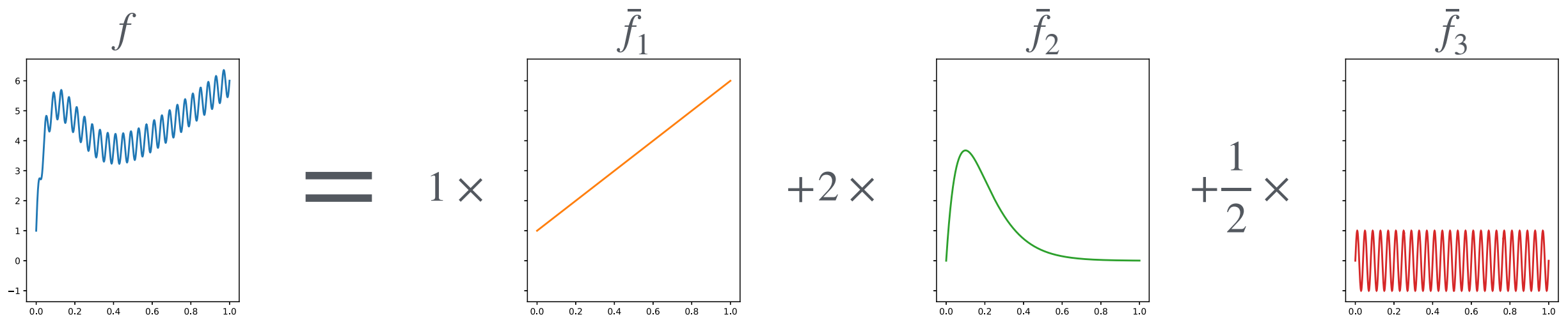
Dictionary learning

Core idea: Representing any object $X \in \mathcal{X}$ by its decomposition $\alpha \in \mathbb{R}^D$ in a dictionary of D objects $\{\bar{X}_i\} \in \mathcal{X}^D$ with a composition function $\mathbf{B}$, i.e. $X \approx \mathbf{B}(\alpha, \{\bar{X}_i\})$.

Training: Given a set of N objects $\{X_j\}$, the dictionary and the encoding vectors are typically learned by minimizing a reconstruction error :

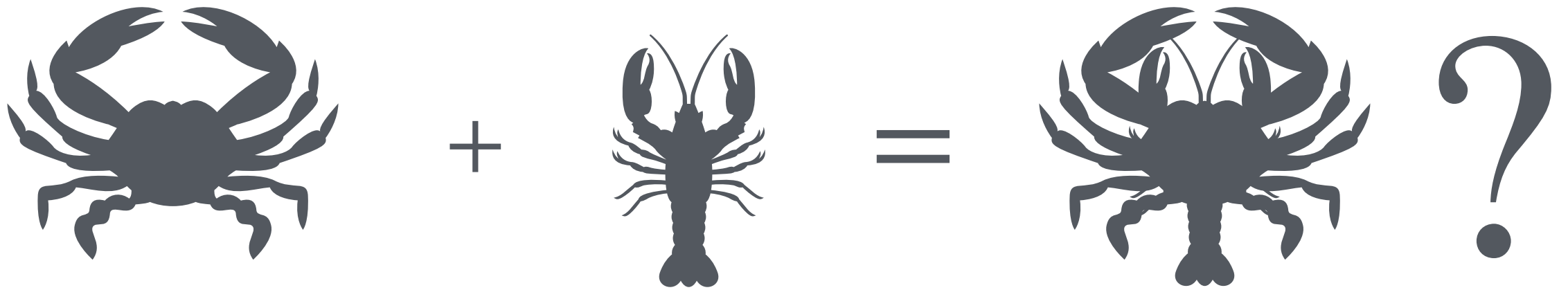
$$\min_{\{\alpha_j\}_{j \in [N]}, \{\bar{X}_i\}_{i \in [D]}} \sum_{j \in [N]} \mathcal{L}(X_j, \mathbf{B}(\alpha_j, \{\bar{X}_i\}))$$

Example on functions:

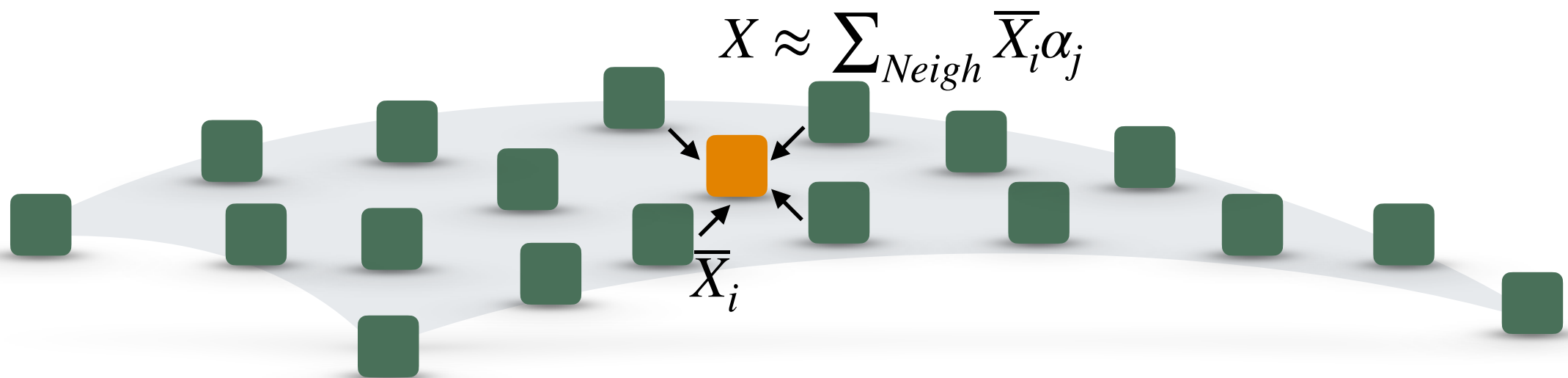


$$\min_{\Lambda, \mathbf{D}} \|\mathbf{X} - \mathbf{D}\Lambda\|_2^2 \quad \text{with} \quad \mathbf{X}_{ij} = f_j(t_i), \mathbf{D}_{ik} = \bar{f}_k(t_i) \text{ and } \Lambda_{k,j} = \alpha_{kj}$$

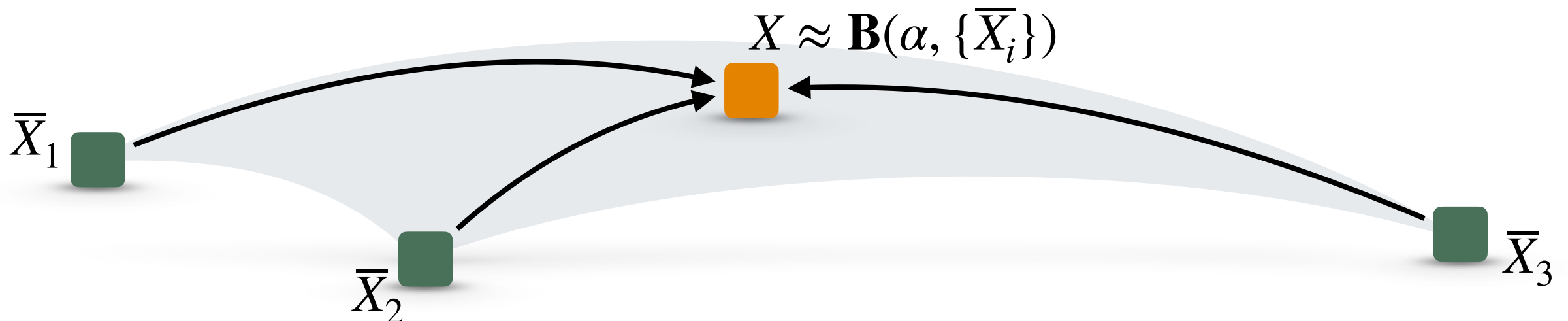
Dictionary learning in non-Euclidean space



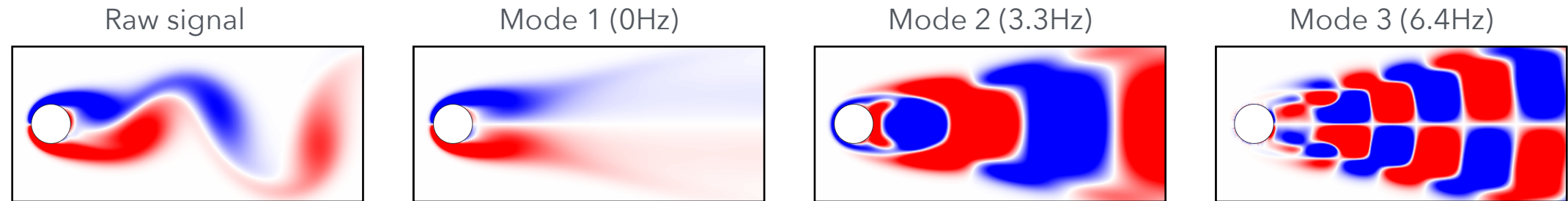
1st strategy: Overcomplete (Large dictionary with sparse coding).



2nd strategy: Incorporating an apriori of the dataset manifold in modeling.



Dictionary learning on operator of DS



Geometry of spectral decomposition: Let $\mathcal{H}$ be a p -dimensional Hilbert space and $\mathcal{S}_r(\mathcal{H})$ be the set of non-defective operators with rank at most r and simple spectrum. For any $\mathbf{A} \in \mathcal{S}_r(\mathcal{H})$:

$$\mathbf{A} = \mathbf{R} \text{Diag}(\Lambda) \mathbf{L}^\dagger \quad \text{s.t.} \quad \mathbf{L}^\dagger \mathbf{R} = \mathbf{I}_r,$$

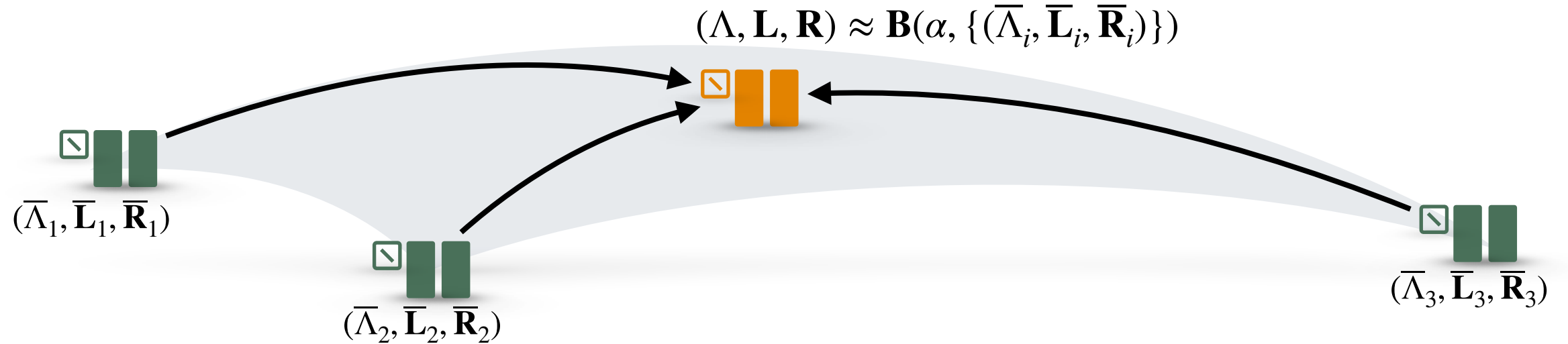
with eigenvalues $\Lambda \in \mathbb{C}^r$ and left/right eigenvectors $(\mathbf{L}, \mathbf{R}) \in (\mathbb{C}^{p \times r})^2$.

⚠ A spectral decomposition uniquely defines an operator up to rescaling of its eigenvectors. Hence, $\mathcal{S}_r(\mathcal{H})$ can be identified with:

$$\mathcal{M} = \mathcal{N} / (\mathbb{C}^*)^r \quad \text{s.t.} \quad \mathcal{N} = \{(\Lambda, \mathbf{L}, \mathbf{R}) \mid \mathbf{L}^\dagger \mathbf{R} = \mathbf{I}_r\}$$

Proposition: $(\mathcal{M}, g)$ is a Riemannian quotient manifold of $(\mathcal{N}, g)$ where g is the stable Riemannian metric.

Dictionary learning on operator of DS



Objective: Given a set of N operators with decomposition $\{(\Lambda_j, \mathbf{L}_j, \mathbf{R}_j)\}$, the dictionary of D atoms and the encoding vectors are learned with:

$$\min_{\{\alpha_j\}, \{(\bar{\Lambda}_i, \bar{\mathbf{L}}_i, \bar{\mathbf{R}}_i)\}} \sum_{j \in [N]} d_{\mathcal{S}}^2 \left((\Lambda_j, \mathbf{L}_j, \mathbf{R}_j), \mathbf{B}(\alpha_j, \{(\bar{\Lambda}_i, \bar{\mathbf{L}}_i, \bar{\mathbf{R}}_i)\}) \right)$$

with the composition function:

$$\begin{aligned} \mathbf{B}(\alpha_j, \{(\bar{\Lambda}_i, \bar{\mathbf{L}}_i, \bar{\mathbf{R}}_i)\}) &= \mathbf{P}_{\mathcal{N}} \left(\sum_i \alpha_{ji} \bar{\Lambda}_i, \sum_i \alpha_{ji} \bar{\mathbf{L}}_i, \sum_i \alpha_{ji} \bar{\mathbf{R}}_i \right) \\ \text{s.t.} \quad \mathbf{P}_{\mathcal{N}}(\Lambda, \mathbf{L}, \mathbf{R}) &= (\Lambda, \arg \min_{\tilde{\mathbf{L}}} \|\tilde{\mathbf{L}} - \mathbf{L}\|_F, \mathbf{R}) \end{aligned}$$

Optimization: Stochastic inexact block-coordinate strategy

1. Coefficient estimation: complete gradient descent
2. Dictionary update: One Riemannian gradient descent iteration

Spectral-Grassmann Optimal transport [Ger25]

Spectral decomposition: For any $A \in \mathcal{S}_r(\mathcal{H})$,

$$Af = \sum_{j \in [r]} \lambda_j \langle g_j, f \rangle f_j = \sum_{j \in [r]} \lambda_j P_j f \quad \text{s.t.} \quad P_j f = \langle g_j, f \rangle f_j$$

with $A := (\Lambda, \mathbf{L}, \mathbf{R})$ then $\lambda_i := \Lambda_i$ and $P_i = \mathbf{R}_i \mathbf{L}_i^\dagger$

Distributional embedding: The operator A can be represented as a probability distribution over the product space of eigenvalues and projectors:

$$\mu(A) \triangleq \sum_{i \in [r]} \frac{1}{r} \delta_{(\lambda_i, P_i)} .$$

Metric between operator: Given two operator A and A' of rank r and r' , $d_{\mathcal{S}}$ verifies:

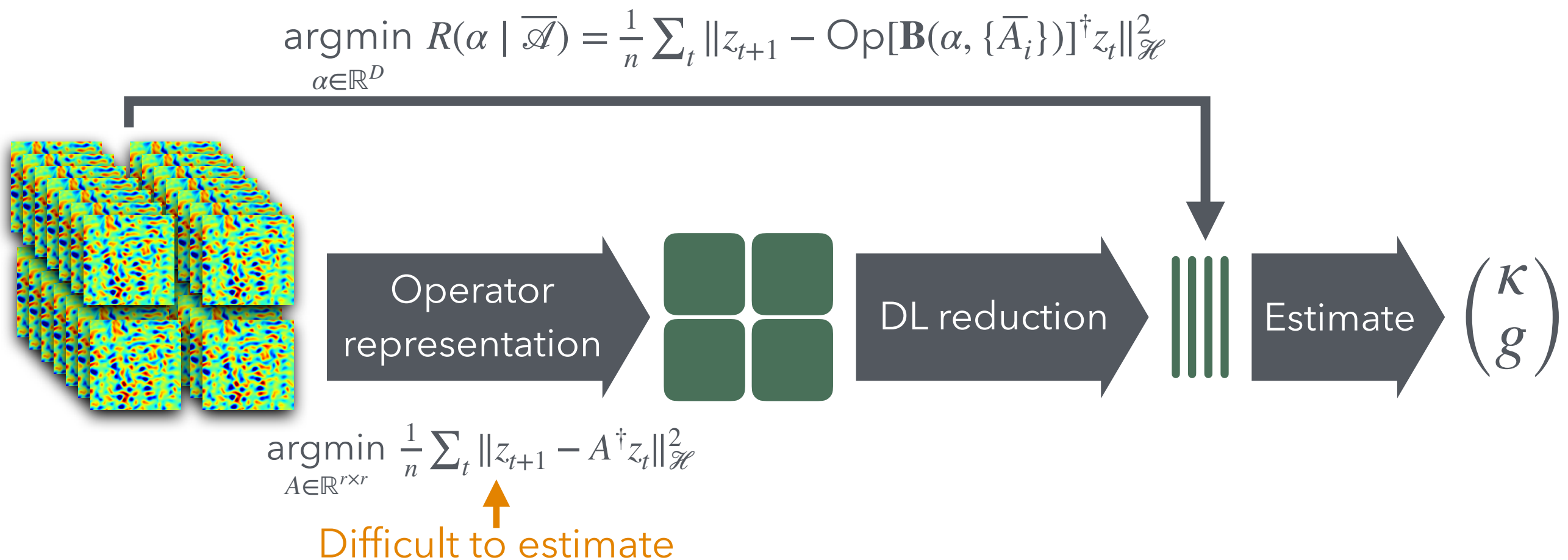
$$d_{\mathcal{S}}(A, A')^2 = \min_{\mathbf{P} \in \Pi} \langle \mathbf{C}, \mathbf{P} \rangle_F ,$$

where $\Pi = \{ \mathbf{P} \in \mathbb{R}_+^{r \times r'} \mid \mathbf{P} \mathbf{1}_{r'} = \mathbf{1}_r, \mathbf{P}^\top \mathbf{1}_r = \mathbf{1}_{r'} \}$, and

$$\mathbf{C}_{ij} = \eta |\lambda_i - \lambda'_j|^2 - (1 - \eta) \log(|\langle P_i, P'_j \rangle| / (\|P_i\| \|P'_j\|) + \epsilon)$$

Dictionary-based operator estimates

Problematic: How to estimate an operator from a short trajectory ?

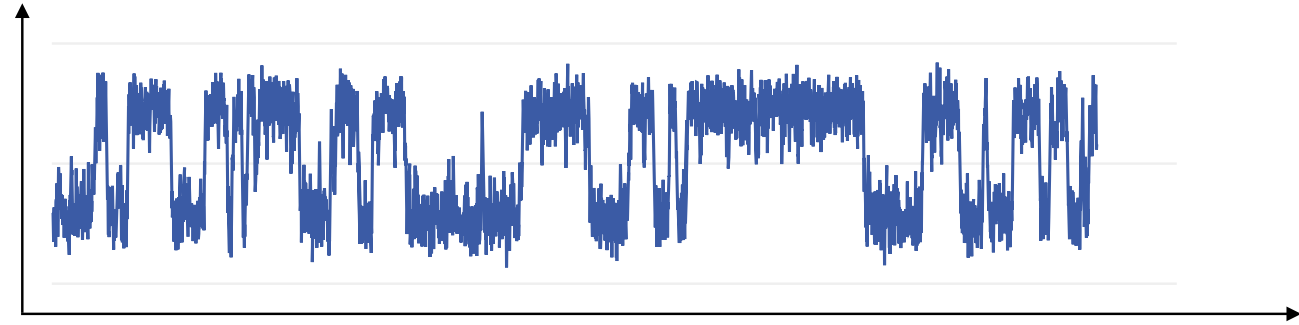
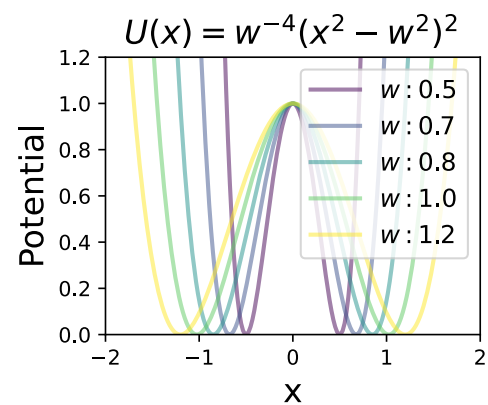


Theorem (Oracle inequality, i.i.d. case): Let $\overline{\mathcal{A}}$ be a dictionary with d atoms and $\mathcal{A}^*$ be its optimal counterpart. Suppose $\{X_t\}_{t \in [n]}$ are i.i.d. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$:

$$R(\hat{\alpha} \mid \overline{\mathcal{A}}) - \inf_{\alpha} R(\alpha \mid \mathcal{A}^*) \leq \text{bias}(\overline{\mathcal{A}}) + \sqrt{\text{Const}(\overline{\mathcal{A}}, \delta, \log(n))/n}$$

Experimental results

Langevin dynamic

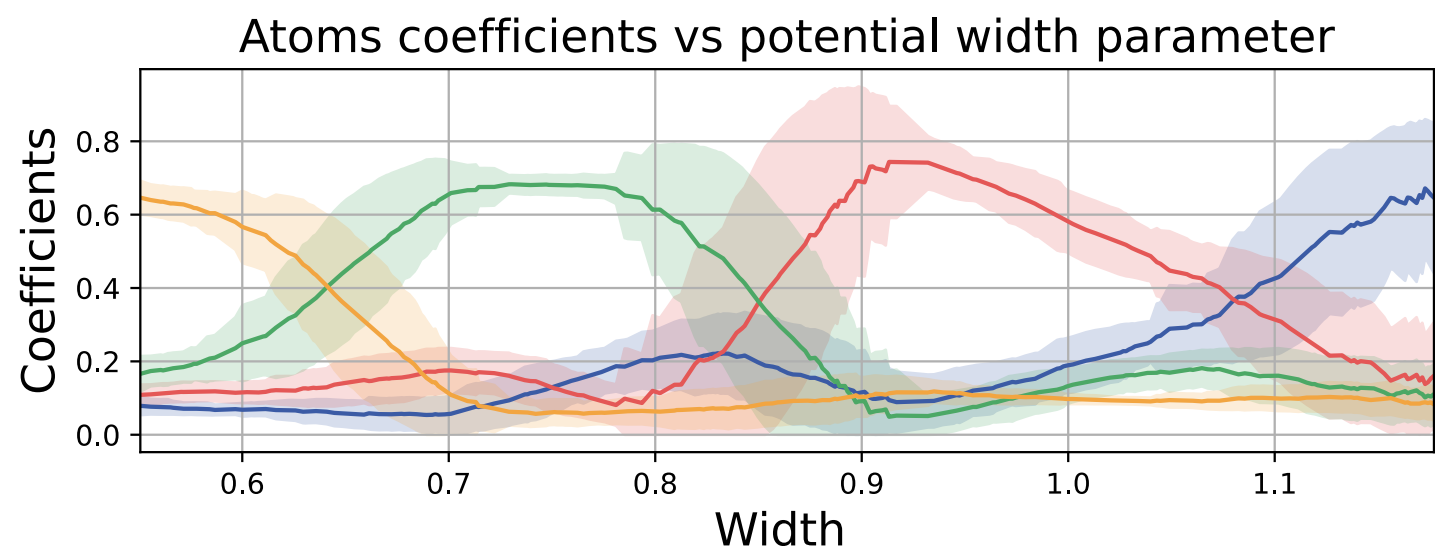
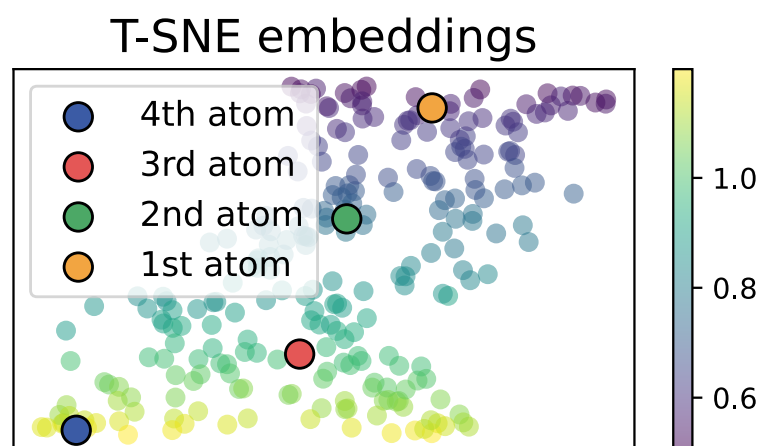


Modeling: (Damped Langevin)

$$dX_t = -\nabla U_w(X_t) + \sqrt{2\sigma}dB_t \text{ with } U_w(x) = w^{-4}(x^2 - w^2)^2.$$

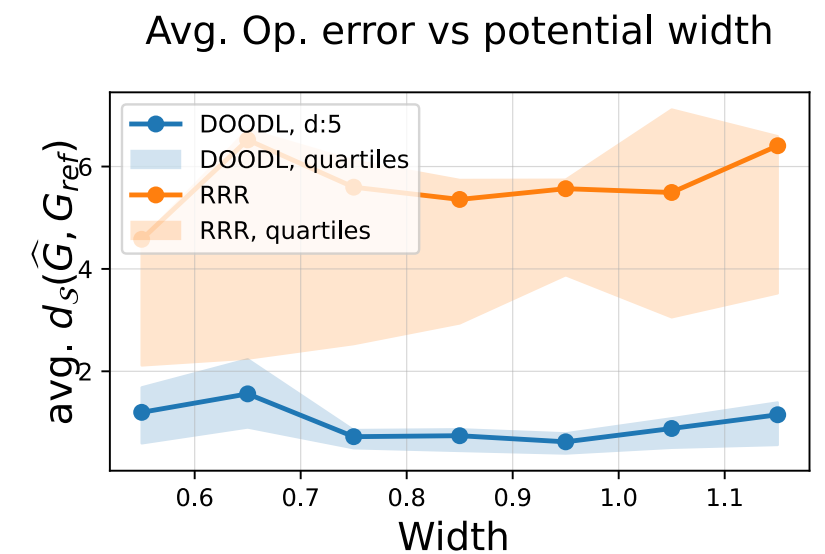
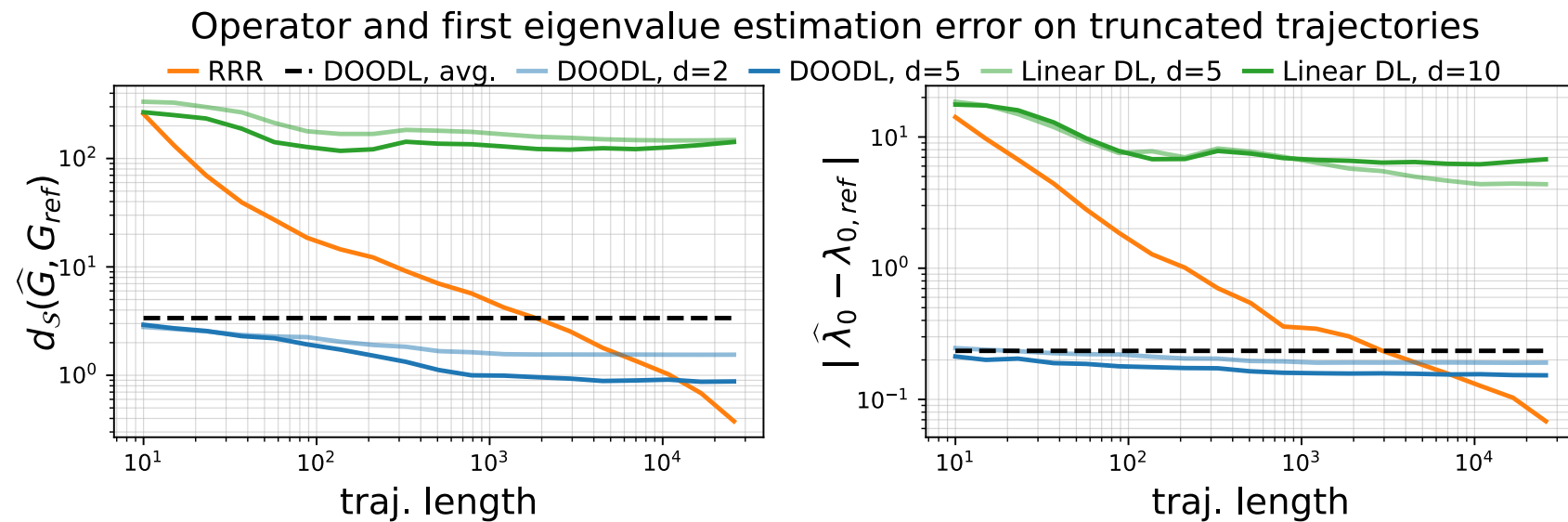
- 256/256 train/test trajectories with 40k samples each and $w \sim \text{Uniform}([0.5, 1.2])$
- Operator estimation: Reduced Rank Regression (RRR) [Kos22]
- Feature map: 400 Random Fourier Features (RFF) on 50 samples sliding windows.

Dictionary learning on train set:

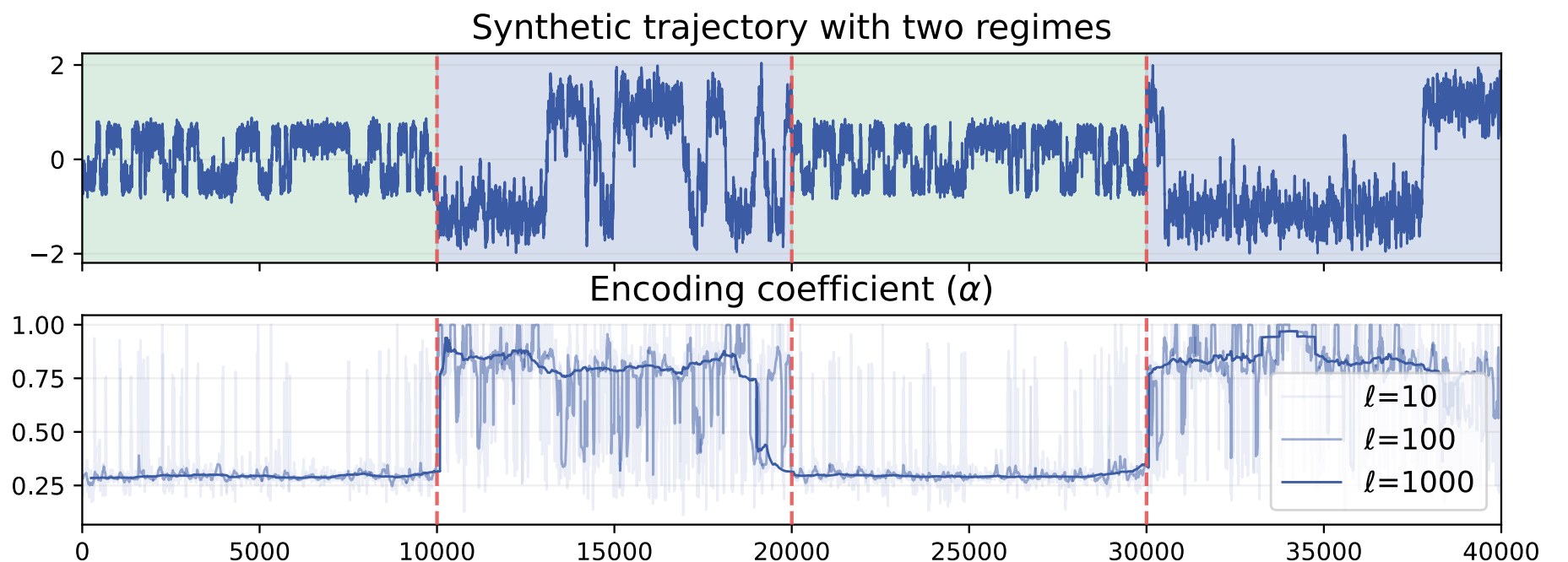
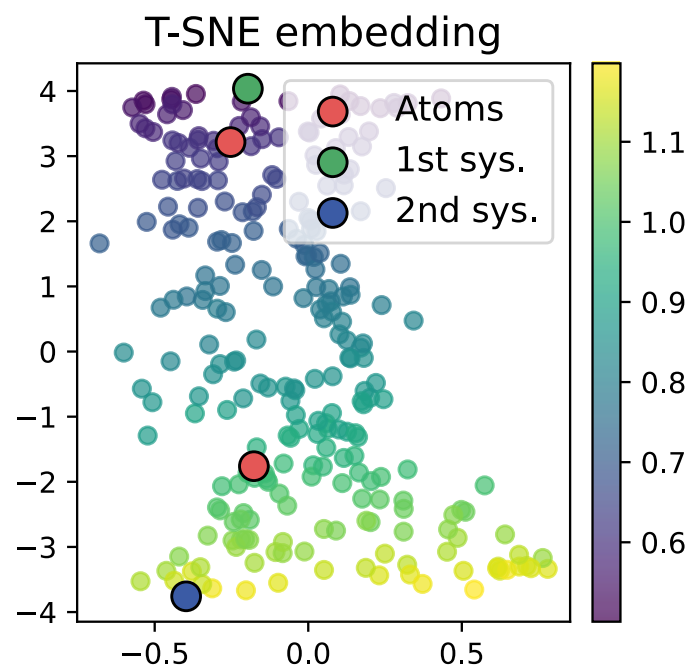


Langevin dynamic

Short trajectory operator estimation:



Switching dynamics estimation:



Plasma dataset generation

Plasma PDE model:

$$\begin{cases} \partial_t n + \kappa \partial_y \phi + [\phi, n] - D_n \Delta_{\perp} n = -\sigma_n n + \sigma_{n\phi} \phi, \\ \partial_t W + g \partial_y n + [\phi, W] - D_{\phi} \Delta_{\perp} W = -\sigma_{\phi n} n + \sigma_{\phi\phi} \phi \\ W = \Delta_{\perp} \phi \end{cases}$$

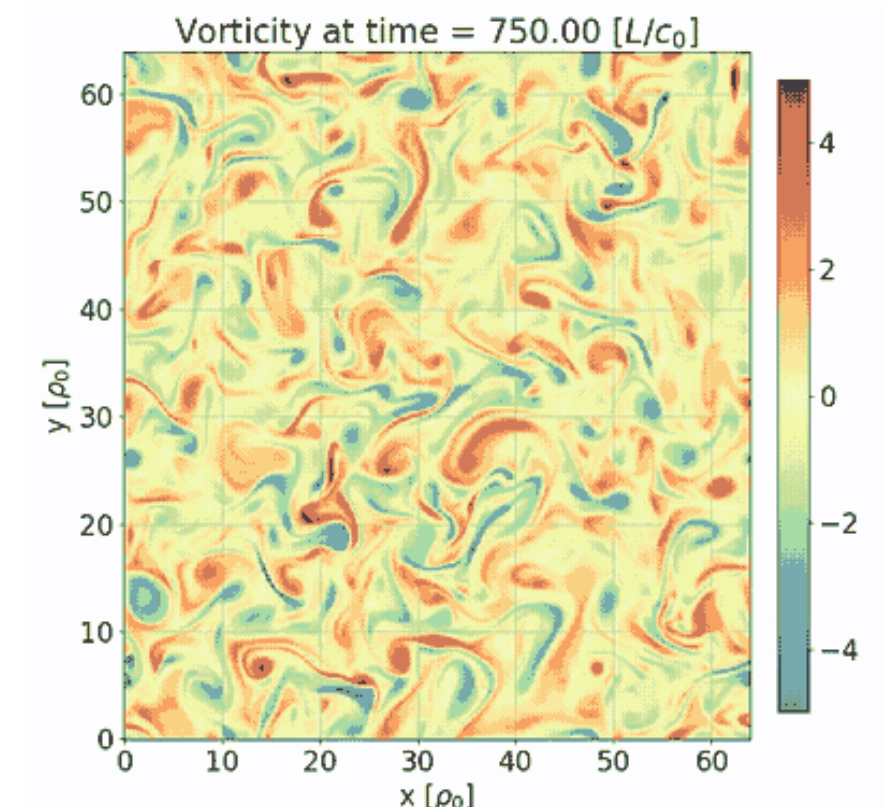
with n the particle density, ϕ the electric potential and W the vorticity.

Goal: Estimation of parameters κ and g , even from short trajectories.

Simulator: Tokam2D [Var23]

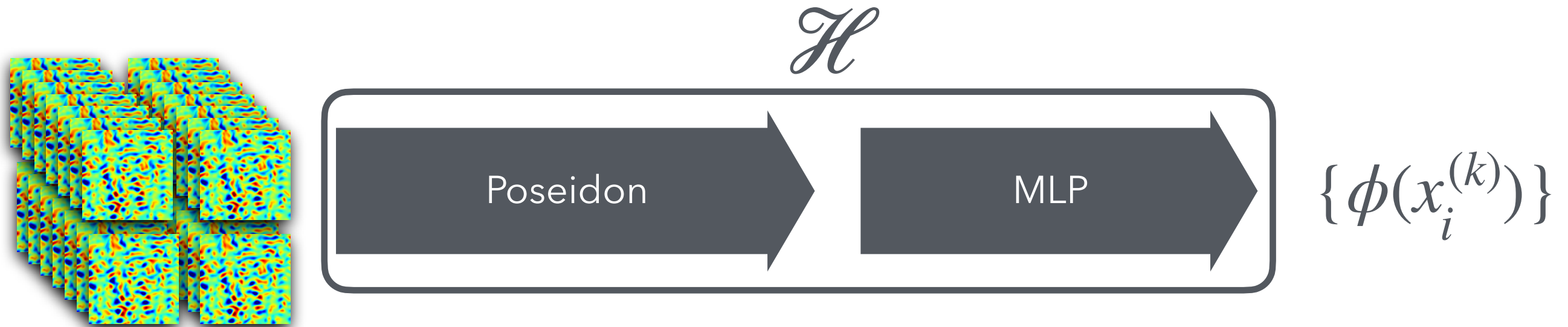
Data generation:

- 320/80 train/test trajectories.
- 128x128 grid, $\Delta t = 0.05s$, $n_{\text{samples}} = 4093$
- $(\kappa, g) \sim \text{Uniform}([1, 3.5] \times [0, 0.5])$
- Laplace transform operator estimator [Kos24]



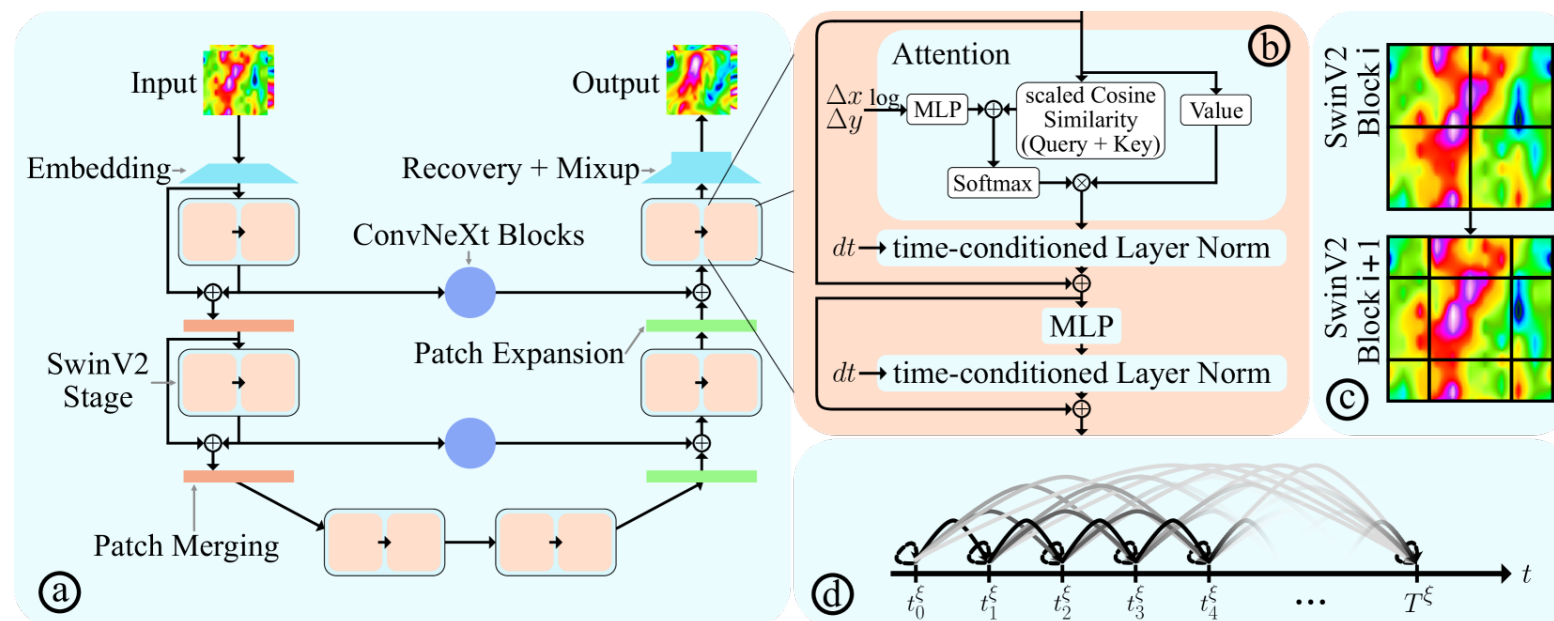
Observable space

Context: Learning a function space $\mathcal{H}$ common for the proper estimation of the whole family of systems $\{A^{(k)}\}_{k \in [K]}$.



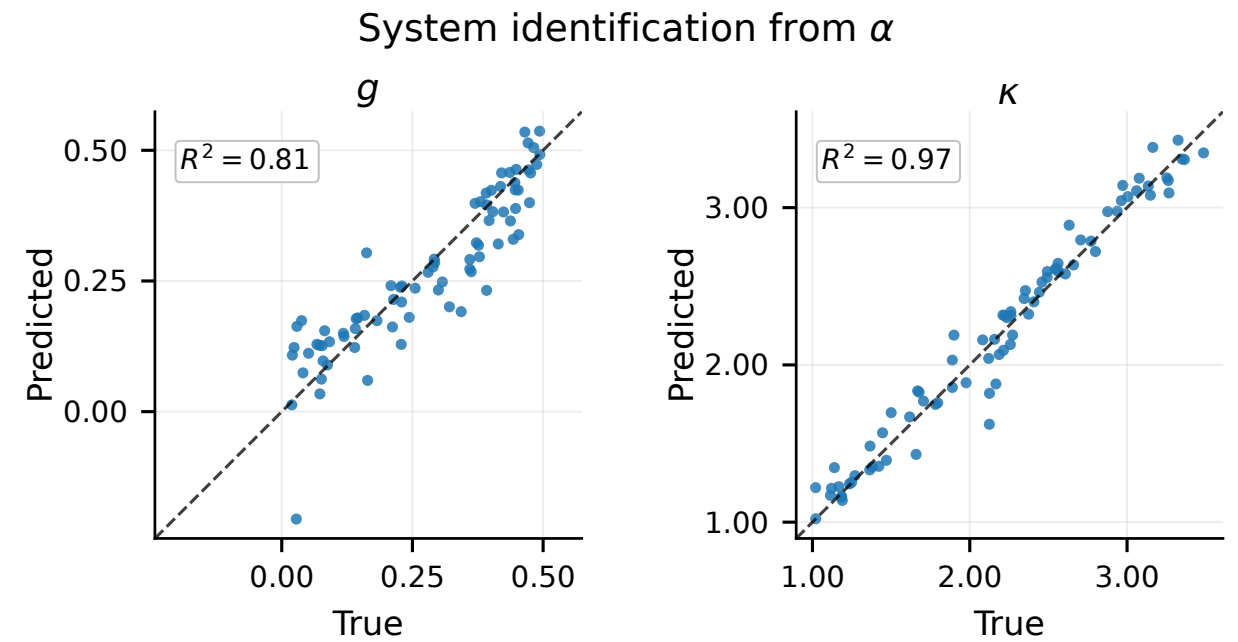
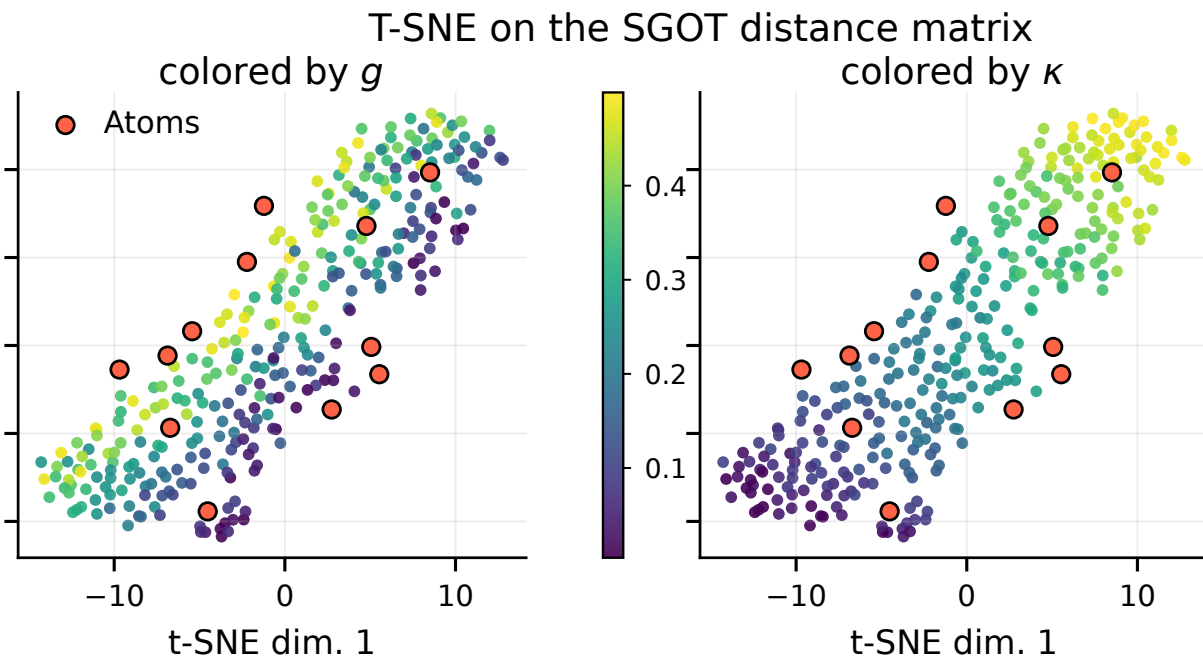
Training: Poseidon frozen, Fine tuning the MLP following [Kos24].

Poseidon: Foundation model for fluid dynamics (transformer based).

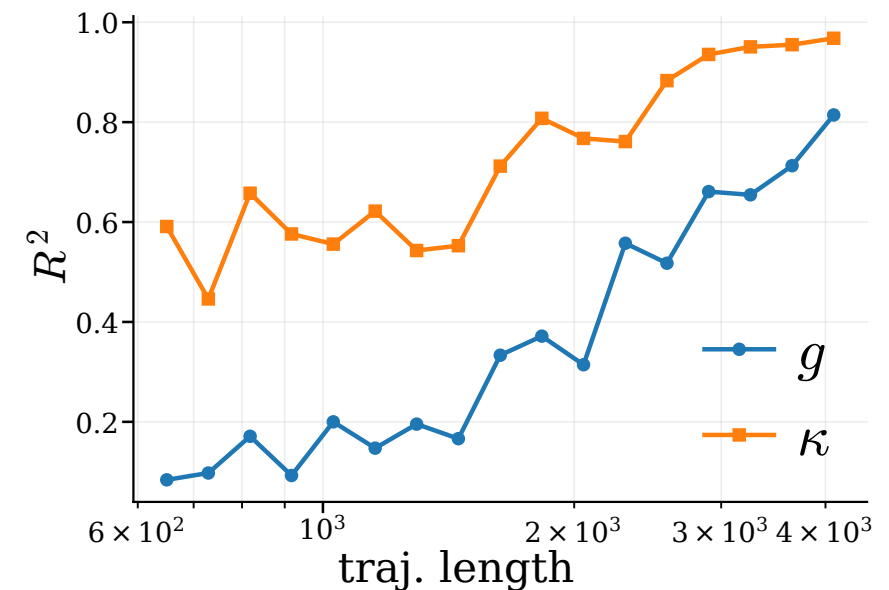
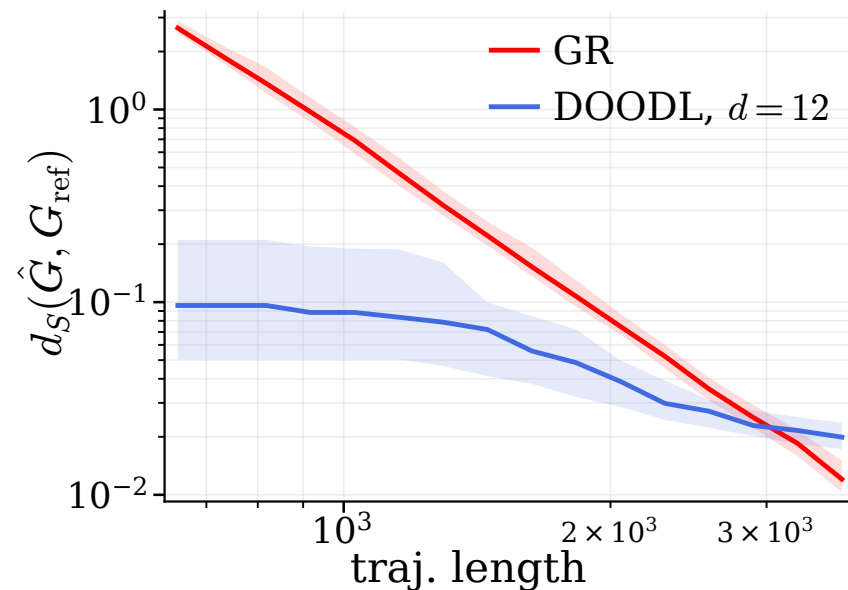


Dictionary learning & Parameter estimation

Full trajectory:



Small to full trajectory:



Thank you !
Any questions ?